Chapter 2

Vector Spaces and the Four Fundamental Subspaces

2.1 Spaces and Subspaces

A **vector space** is a set of vectors equipped with vector addition and scalar multiplication satisfying the vector-space axioms. Scalars may be real or complex. In this chapter, we work over the real numbers with the usual operations. In particular, a vector space is closed under these operations:

- If $\mathbf{x}, \mathbf{y} \in V$, then $\mathbf{x} + \mathbf{y} \in V$.
- If $\mathbf{x} \in V$ and $\alpha \in \mathbb{R}$, then $\alpha\mathbf{x} \in V$.

Consequently, every linear combination $\alpha\mathbf{x} + \beta\mathbf{y}$ of vectors in V also belongs to V. Closure alone is not a complete definition of an abstract vector space; the other axioms include associativity, commutativity of addition, distributivity, and the existence of a zero vector and additive inverses.

For example,

$$\mathbb{R}^n = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} : a_i \in \mathbb{R},\ i = 1, \ldots, n \right\}. \tag{2.1}$$

The set $\mathbb{R}^n$ consists of all vectors with n real components. With the usual operations, it is a vector space. The familiar xy-plane is $\mathbb{R}^2$, and three-dimensional coordinate space is $\mathbb{R}^3$.

2.1.1 Subspaces

Definition: A nonempty subset $S \subseteq V$ is a **subspace** of V if it is a vector space under the operations inherited from V. Because the remaining axioms are inherited, it suffices to check closure under addition and scalar multiplication.

A line through the origin is a subspace of $\mathbb{R}^2$. A line that does not pass through the origin is not a subspace. Every subspace contains the zero vector, since $0\mathbf{x} = \mathbf{0}$ for any vector $\mathbf{x}$ in it. Containing the origin is necessary, but is not sufficient by itself.

Consider

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{pmatrix}. \tag{2.2}$$

The columns of A belong to $\mathbb{R}^3$. The smallest subspace containing both columns consists of all their linear combinations. This is the **column space**, denoted by $C(A)$.

Now consider

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{pmatrix}. \tag{2.3}$$

Its column space is a subspace of $\mathbb{R}^4$:

$$A\mathbf{x} = x_1 \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} + x_3 \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 + 2x_3 \\ 2x_1 + x_2 + 3x_3 \\ 3x_1 + x_2 + 4x_3 \\ 4x_1 + x_2 + 5x_3 \end{pmatrix}. \tag{2.4}$$

Three vectors cannot span all of $\mathbb{R}^4$. In this example, the third column is the sum of the first two, and the first two are independent. Thus $C(A)$ is a two-dimensional subspace of $\mathbb{R}^4$.

Which vectors $\mathbf{b}$ make $A\mathbf{x} = \mathbf{b}$ solvable? Precisely those in the column space:

$$A\mathbf{x} = \mathbf{b} \text{ is solvable} \quad \Longleftrightarrow \quad \mathbf{b} \in C(A). \tag{2.5}$$

For $\mathbf{b} = \mathbf{0}$, one solution is $\mathbf{x} = \mathbf{0}$. For $\mathbf{b} = A_{*1}$, one solution is $(1, 0, 0)^T$. For $\mathbf{b} = A_{*2}$, both $(0, 1, 0)^T$ and $(-1, 0, 1)^T$ are solutions.

2.1.2 Nullspace

Definition: The **nullspace** of an $m \times n$ matrix A is

$$N(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\}. \tag{2.6}$$

For the matrix in (2.3), the relation $A_{*1} + A_{*2} - A_{*3} = \mathbf{0}$ gives

$$A \begin{pmatrix} c \\ c \\ -c \end{pmatrix} = \mathbf{0}, \qquad c \in \mathbb{R}. \tag{2.7}$$

These are all the solutions: independence of the first two columns implies $x_1 + x_3 = 0$ and $x_2 + x_3 = 0$. The nullspace is therefore the line through the origin in the direction $(1, 1, -1)^T$.

The nullspace is always a subspace. It contains $\mathbf{0}$, and if $A\mathbf{u} = A\mathbf{v} = \mathbf{0}$, then

$$A(\alpha\mathbf{u} + \beta\mathbf{v}) = \alpha A\mathbf{u} + \beta A\mathbf{v} = \mathbf{0}. \tag{2.8}$$

By contrast, the solutions of $A\mathbf{x} = A_{*1}$ do not form a vector space, since $\mathbf{0}$ is not a solution. Their complete description is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \qquad c \in \mathbb{R}. \tag{2.9}$$

This is a line in $\mathbb{R}^3$ that does not pass through the origin. It includes both $(1, 0, 0)^T$ and $(0, -1, 1)^T$.

2.2 Solving Homogeneous Systems and Reduced Row Echelon Form

For larger matrices, we need a systematic way to find the nullspace and column space. Gaussian elimination extends to rectangular matrices, producing **row echelon form** rather than necessarily an upper triangular matrix. In echelon form, zero rows are at the bottom, each nonzero row has a leading nonzero entry called a pivot, and each pivot lies to the right of the pivot in the preceding row.

Consider

$$A = \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 \\ 3 & 6 & 1 & 4 \end{pmatrix}. \tag{2.10}$$

To solve $A\mathbf{x} = \mathbf{0}$, we can work directly on A, since the right-hand side remains zero. First use $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - 3R_1$:

$$A \longrightarrow \begin{pmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & -5 & -5 \end{pmatrix}. \tag{2.11}$$

Column 2 has no available pivot below the first row, so the second pivot is -3 in column 3. Using $R_3 \leftarrow R_3 - \frac{5}{3}R_2$ gives

$$U = \begin{pmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & -3 & -3 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.12}$$

```
A = [1 2 2 3;2 4 1 3;3 6 1 4];
U = A;
U(2,:) = U(2,:) - 2*U(1,:);
U(3,:) = U(3,:) - 3*U(1,:);
U(3,:) = U(3,:) - (5/3)*U(2,:)
```

The **rank** of A is the number of pivots. Here $r = \operatorname{rank}(A) = 2$. The pivot columns are columns 1 and 3; the other columns are called free columns. The corresponding variables x_1 and x_3 are **basic variables**, while x_2 and x_4 are **free variables**.

The nonzero equations are

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 3x_4 &= 0, \\ -3x_3 - 3x_4 &= 0. \end{aligned} \tag{2.13}$$

We may assign any real values to x_2 and x_4. Back substitution gives $x_3 = -x_4$ and $x_1 = -2x_2 - x_4$, so

$$\mathbf{x} = x_2 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix}. \tag{2.14}$$

The two vectors are independent, so $N(A)$ is a two-dimensional plane through the origin in $\mathbb{R}^4$.

In general, an $m \times n$ matrix of rank r has $n - r$ free variables. Setting one free variable to 1 and the others to 0 in turn produces $n - r$ **special solutions**, which form a basis for the nullspace. Thus

$$\dim N(A) = n - r. \tag{2.15}$$

2.2.1 Reduced Row Echelon Form

Gauss–Jordan elimination makes each pivot equal to 1 and eliminates every other entry in its column. The result is the unique **reduced row echelon form** (rref) of the matrix. Continuing the example, divide row 2 by -3 and subtract twice the new row 2 from row 1:

$$U \longrightarrow \begin{pmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \longrightarrow R = \operatorname{rref}(A) = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.16}$$

```
A = [1 2 2 3;2 4 1 3;3 6 1 4];
R = rref(A)
```

The rref reveals the rank, pivot columns, basic variables, and nullspace equations. It also reveals the dependence relations among the original columns. Here

$$A_{*2} = 2A_{*1}, \qquad A_{*4} = A_{*1} + A_{*3}. \tag{2.17}$$

Row operations preserve these relations because they multiply every column by the same invertible matrix. A basis for $C(A)$ must be taken from the pivot columns of the *original* matrix, not from the reduced matrix.

2.2.2 A Block Formula for the Nullspace

If the pivot columns are placed first by a column permutation Q, the reduced matrix has the form

$$RQ = \begin{pmatrix} I_r & F \\ 0 & 0 \end{pmatrix}, \qquad F \in \mathbb{R}^{r \times (n-r)}. \tag{2.18}$$

Here Q is a permutation matrix and $\mathbf{x} = Q\mathbf{z}$, so $\mathbf{z}$ lists the variables in the reordered column order. Splitting $\mathbf{z}$ into basic and free components gives $\mathbf{z}_{\text{basic}} = -F\mathbf{z}_{\text{free}}$. Therefore a nullspace basis matrix in the original variable order is

$$N = Q\begin{pmatrix} -F \\ I_{n-r} \end{pmatrix}, \qquad AN = 0. \tag{2.19}$$

The identity matrix has $n - r$ rows and columns, matching the number of columns of F. The factor Q restores the original variable order and may be omitted only when no column permutation is needed.

For (2.16), interchanging columns 2 and 3 gives

$$RQ = \begin{pmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}. \tag{2.20}$$

Restoring the original order (x_1, x_2, x_3, x_4) yields

$$N = \begin{pmatrix} -2 & -1 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}, \tag{2.21}$$

whose columns are the special solutions in (2.14). MATLAB's `null(A)` generally returns an orthonormal basis computed numerically, so its columns may differ from these special solutions while spanning the same space.

Example: Consider

$$B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 2 & 6 & 8 \\ 2 & 8 & 10 \end{pmatrix}, \qquad \operatorname{rref}(B) = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{2.22}$$

The first two columns are independent, and the third is their sum. There are two pivots and one free variable. Thus

$$\mathbf{x} = x_3\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}, \qquad N = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}. \tag{2.23}$$

```
B = [1 2 3;2 4 6;2 6 8;2 8 10];
rref(B)
```

□

2.3 Nonhomogeneous Systems and Solvability

Return to the matrix in (2.10), now with a general right-hand side. Applying the same row operations to the augmented matrix, and multiplying the last row by 3 for convenience, gives

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & (2b_2 - b_1)/3 \\ 0 & 0 & 1 & 1 & (2b_1 - b_2)/3 \\ 0 & 0 & 0 & 0 & b_1 - 5b_2 + 3b_3 \end{array}\right). \tag{2.24}$$

```
A = [1 2 2 3;2 4 1 3;3 6 1 4];
syms b1 b2 b3 real
b = [b1;b2;b3];
M = [A b];
M(2,:) = M(2,:) - 2*M(1,:);
M(3,:) = M(3,:) - 3*M(1,:);
M(3,:) = M(3,:) - (5/3)*M(2,:);
M(2,:) = M(2,:)/(-3);
M(1,:) = M(1,:) - 2*M(2,:);
M(3,:) = 3*M(3,:)
```

The last row gives the **solvability condition**

$$b_1 - 5b_2 + 3b_3 = 0. \tag{2.25}$$

If this condition fails, the last row is a contradiction. Equivalently, the augmented right-hand side introduces an additional pivot, so $\mathbf{b} \notin C(A)$.

For $\mathbf{b} = (-1, 1, 2)^T$, the condition holds, and (2.24) becomes

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right). \tag{2.26}$$

Set the free variables to zero to obtain a **particular solution**, $\mathbf{x}_p = (1, 0, -1, 0)^T$. Every solution is this particular solution plus a nullspace vector:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_n = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix}. \tag{2.27}$$

The solution set is an affine plane in $\mathbb{R}^4$: a translation of $N(A)$ that does not pass through the origin. In general, for any consistent system,

$$\{\mathbf{x} : A\mathbf{x} = \mathbf{b}\} = \mathbf{x}_p + N(A). \tag{2.28}$$

Example: A system with more equations than unknowns is

$$\begin{aligned} 2x_1 + 4x_2 + 6x_3 &= 2, \\ x_1 + 2x_2 + 3x_3 &= 1, \\ x_1 + x_3 &= -3, \\ 2x_1 + 4x_2 &= 8. \end{aligned} \tag{2.29}$$

Its augmented matrix reduces to

$$\left(\begin{array}{ccc|c} 2 & 4 & 6 & 2 \\ 1 & 2 & 3 & 1 \\ 1 & 0 & 1 & -3 \\ 2 & 4 & 0 & 8 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right). \tag{2.30}$$

```
A = [2 4 6;1 2 3;1 0 1;2 4 0];
b = [2;1;-3;8];
rref([A b])
```

The system is consistent and $\operatorname{rank}(A) = 3$, equal to the number of unknowns. There are no free variables, so the unique solution is $\mathbf{x} = (-2, 3, -1)^T$.

If the first right-hand side is changed from 2 to 1, the first two equations contradict each other: their left-hand sides differ by a factor of 2, while their right-hand sides do not. The resulting augmented matrix has rank 4 and rref I_4, whose last row represents $0 = 1$. □

2.3.1 Rank and the Number of Solutions

For any $m \times n$ matrix of rank r,

$$r \leq \min(m, n). \tag{2.31}$$

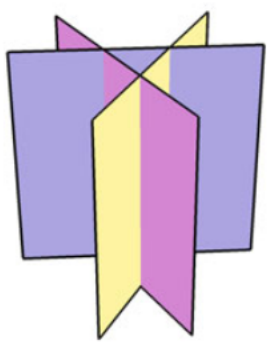
There are three possibilities for $A\mathbf{x} = \mathbf{b}$:

- **Exactly one solution**: The system is consistent and $r = n$ (full column rank). There are no free variables and $N(A) = \{\mathbf{0}\}$.
- **Infinitely many solutions**: The system is consistent and $r < n$. There are $n - r$ free variables.
- **No solution**: The system is inconsistent. Elimination produces a zero coefficient row with a nonzero right-hand side.

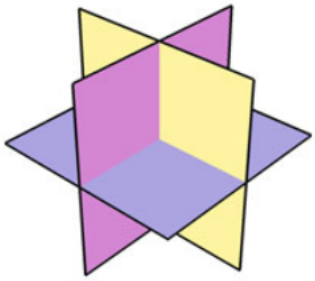
If $r = m$ (full row rank), there is a pivot in every row, so the system is solvable for every $\mathbf{b} \in \mathbb{R}^m$. For example,

$$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 1 & 1 \end{pmatrix} \tag{2.32}$$

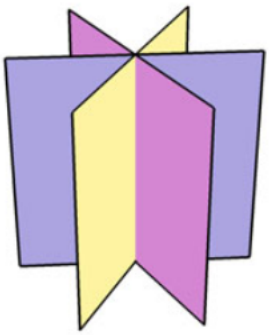
has rank 2 and is solvable for every right-hand side. Since $n - r = 2$, every such system has infinitely many solutions. If $r = m = n$, the matrix is square and invertible, its rref is I_n, and every right-hand side has exactly one solution.



No Solution



Unique Solution



Infinitely Many Solutions

The following table summarizes the possibilities. In the displayed reduced forms, pivot columns are understood to have been placed first when necessary.

Rank condition	**Reduced form**	**Solutions of $A\mathbf{x} = \mathbf{b}$**
$r = m = n$	I_n	Exactly one for every $\mathbf{b}$.
$r = n < m$	$\begin{pmatrix} I_n \\ 0 \end{pmatrix}$	Zero or one, depending on consistency.
$r = m < n$	$\begin{pmatrix} I_m & F \end{pmatrix}$	Infinitely many for every $\mathbf{b}$; $n - m$ free variables.
$r < m$ and $r < n$	$\begin{pmatrix} I_r & F \\ 0 & 0 \end{pmatrix}$	Zero or infinitely many, depending on consistency.

2.4 Independence, Basis, and Dimension

If $m < n$, a homogeneous system with m equations and n unknowns always has nonzero solutions, since $r \le m < n$ leaves at least one free variable. This observation connects elimination with linear independence.

Definition: Vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ are **linearly independent** if

$$\alpha_1 \mathbf{v}_1 + \cdots + \alpha_n \mathbf{v}_n = \mathbf{0} \quad \Longrightarrow \quad \alpha_1 = \cdots = \alpha_n = 0. \tag{2.33}$$

Otherwise they are **linearly dependent**.

For example, $(1, 1)^T$ and $(2, 2)^T$ are dependent because twice the first vector minus the second is zero. Any set containing the zero vector is dependent: a nonzero coefficient on that vector, with zero coefficients on all others, gives a nontrivial zero combination.

The vectors $(1, 1)^T$ and $(1, 2)^T$ are independent. Adding $(1, -1)^T$ makes the set dependent. To see this systematically, put the vectors into columns:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -1 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}. \tag{2.34}$$

Because there are more columns than rows, $A\mathbf{x} = \mathbf{0}$ has a nonzero solution. For instance, $-3\mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}$.

In general, the columns of A are independent if and only if $N(A) = \{\mathbf{0}\}$, or equivalently $r = n$. There cannot be more than n independent vectors in $\mathbb{R}^n$.

2.4.1 Spanning Sets and Bases

Definition: For $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_r\} \subseteq V$, its **span** is the set

$$\operatorname{span}(S) = \{\alpha_1\mathbf{v}_1 + \cdots + \alpha_r\mathbf{v}_r : \alpha_1, \ldots, \alpha_r \in \mathbb{R}\}. \tag{2.35}$$

The span is a subspace of V. The columns of a matrix span its column space, but a spanning set can include redundant vectors. For example, $(1,0)^T$ and $(1,1)^T$ span $\mathbb{R}^2$; adding $(1,3)^T$ does not enlarge the span because

$$\begin{pmatrix}1\\3\end{pmatrix} = -2\begin{pmatrix}1\\0\end{pmatrix} + 3\begin{pmatrix}1\\1\end{pmatrix}. \tag{2.36}$$

Definition: A **basis** for V is a linearly independent spanning set for V. Both properties are essential: the vectors must span the space and must be independent.

The standard basis for $\mathbb{R}^3$ is

$$\mathbf{e}_1 = \begin{pmatrix}1\\0\\0\end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix}0\\1\\0\end{pmatrix}, \qquad \mathbf{e}_3 = \begin{pmatrix}0\\0\\1\end{pmatrix}. \tag{2.37}$$

There are infinitely many other bases. The vectors $(1,1,2)^T$ and $(2,2,5)^T$ span a plane in $\mathbb{R}^3$. Adding $(3,1,6)^T$ gives a basis for $\mathbb{R}^3$, because the matrix formed from these columns has rank 3.

```
v1 = [1;1;2];
v2 = [2;2;5];
v3 = [3;1;6];
B = [v1 v2 v3];
rank(B)
```

Definition: The **dimension** of a finite-dimensional vector space is the number of vectors in any basis. Every basis for $\mathbb{R}^n$ has exactly n vectors.

Example: Consider

$$A = \begin{pmatrix}1 & 1 & 2 & 2\\ 1 & 2 & 3 & 2\\ 1 & 3 & 4 & 2\end{pmatrix}, \qquad R = \operatorname{rref}(A) = \begin{pmatrix}1 & 0 & 1 & 2\\ 0 & 1 & 1 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}. \tag{2.38}$$

The first two columns are independent, the third is their sum, and the fourth is twice the first. Thus the first two original columns form a basis for $C(A)$, and $\dim C(A) = r = 2$. Other choices of independent columns can also form a basis; the pivot columns provide a systematic choice.

There are two free variables, x_3 and x_4. Solving $R\mathbf{x} = \mathbf{0}$ gives

$$\mathbf{x} = x_3 \begin{pmatrix} -1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} -2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \qquad N = \begin{pmatrix} -1 & -2 \\ -1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2.39}$$

The columns of N form a basis for $N(A)$, so $\dim N(A) = n - r = 2$.

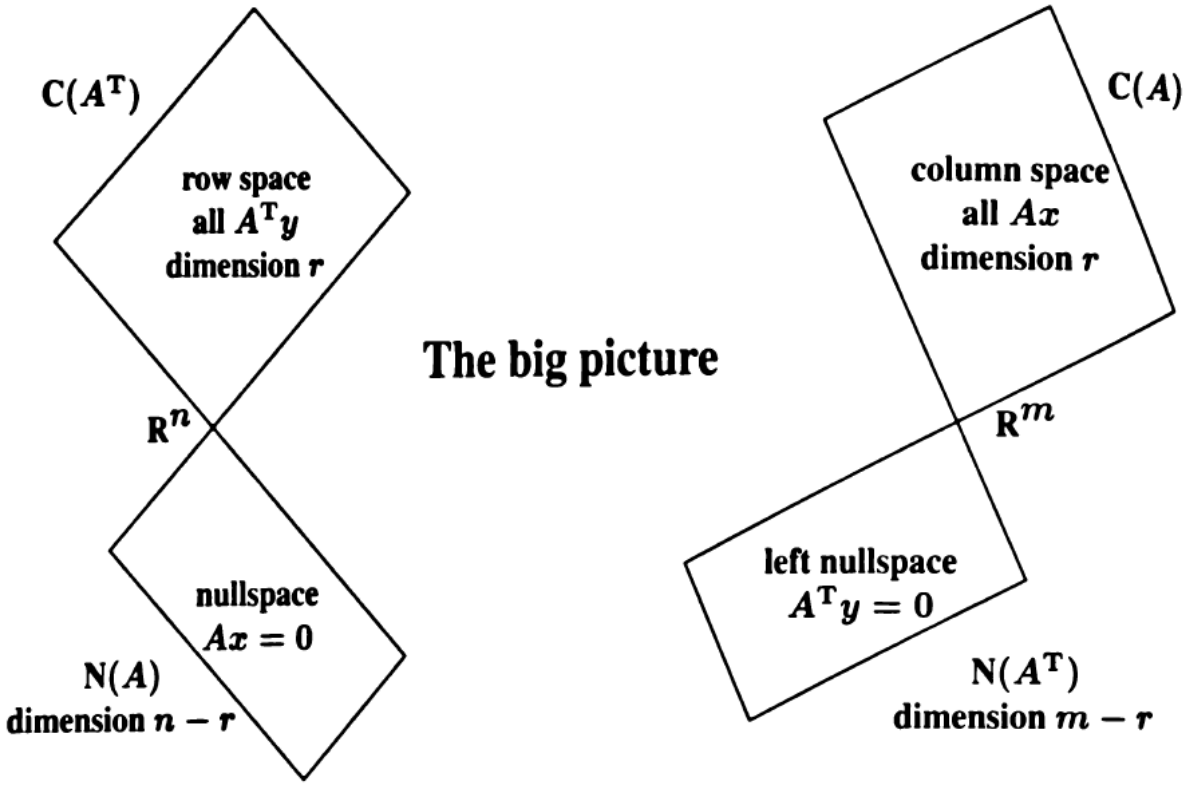
```
A = [1 1 2 2;1 2 3 2;1 3 4 2];
rref(A)
```

□

2.5 The Four Fundamental Subspaces

For a real $m \times n$ matrix A of rank r, the four fundamental subspaces are:

- **Column space** $C(A) \subseteq \mathbb{R}^m$: all linear combinations of the columns of A.
- **Nullspace** $N(A) \subseteq \mathbb{R}^n$: all solutions of $A\mathbf{x} = \mathbf{0}$.
- **Row space** $C(A^T) \subseteq \mathbb{R}^n$: the span of the rows of A, regarded as column vectors, or equivalently the columns of A^T.
- **Left nullspace** $N(A^T) \subseteq \mathbb{R}^m$: all solutions of $A^T\mathbf{y} = \mathbf{0}$.



Subspace	**Lives in**	**A basis**	**Dim.**
$C(A)$	$\mathbb{R}^m$	Pivot columns of the original A	r
$N(A)$	$\mathbb{R}^n$	Special solutions of $A\mathbf{x} = \mathbf{0}$	$n - r$
$C(A^T)$	$\mathbb{R}^n$	Nonzero rows of rref(A), transposed	r
$N(A^T)$	$\mathbb{R}^m$	Special solutions of $A^T\mathbf{y} = \mathbf{0}$	$m - r$

2.5.1 Finding a Basis for the Row Space

We can find a row-space basis directly from A, without reducing A^T. Elementary row operations replace rows by invertible linear combinations, so they preserve the row space. They generally change the column space.

Example: For the matrix in (2.10),

$$A = \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 \\ 3 & 6 & 1 & 4 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.40}$$

Since $C(A^T) = C(R^T)$, the two nonzero rows of R, transposed, form a basis for the row space:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix} \right\}. \tag{2.41}$$

The row space has dimension 2, equal to the rank. Here $C(A) \neq C(R)$: every vector in $C(R)$ has third component zero, while the first column of A does not.

2.5.2 Finding a Basis for the Left Nullspace

The name follows from transposing its defining equation:

$$A^T\mathbf{y} = \mathbf{0} \quad \Longleftrightarrow \quad \mathbf{y}^T A = \mathbf{0}^T. \tag{2.42}$$

To track the row operations, append the identity matrix and perform Gauss–Jordan elimination:

$$[A \mid I_m] \longrightarrow [R \mid G], \qquad GA = R. \tag{2.43}$$

The matrix G is an invertible $m \times m$ product of elementary matrices. If A is square and invertible, $R = I_m$ and $G = A^{-1}$. For a rectangular matrix, G still records the row operations, even though A has no two-sided inverse.

For the present example, reducing the full augmented matrix gives

$$\text{rref}([A \mid I_3]) = \left(\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 0 & -1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 3 & -2 \\ 0 & 0 & 0 & 0 & 1 & -5 & 3 \end{array} \right). \tag{2.44}$$

Thus

$$G = \begin{pmatrix} 0 & -1 & 1 \\ 0 & 3 & -2 \\ 1 & -5 & 3 \end{pmatrix}, \qquad GA = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{2.45}$$

The last row of G produces a zero row when multiplied by A, so

$$\mathbf{y} = \begin{pmatrix} 1 \\ -5 \\ 3 \end{pmatrix}, \qquad A^T\mathbf{y} = \mathbf{0}. \tag{2.46}$$

Since $m - r = 1$, this nonzero vector forms a basis for $N(A^T)$.

```
A = [1 2 2 3;2 4 1 3;3 6 1 4];
EM = rref([A eye(3)]);
R = EM(:,1:4);
G = EM(:,5:7);
G*A
y = G(3,:)';
A'*y
```

In general, the $m - r$ rows of G corresponding to zero rows of R, transposed, form a basis for $N(A^T)$. They are independent because G is invertible.

The same vector explains the earlier solvability condition: if $A\mathbf{x} = \mathbf{b}$, then $\mathbf{y}^T\mathbf{b} = \mathbf{y}^TA\mathbf{x} = 0$, giving $b_1 - 5b_2 + 3b_3 = 0$. □